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No. of Printed Pages : 3

Seat no. _____



SARDAR PATEL UNIVERSITY
BCA (SEMESTER-I) EXAMINATION(NC) 2023
US01FBCA02 Mathematics I

Time: 10:00AM to 1:00PM

Date: 06/12/2023

Total Marks: 70

Q-1 Select the correct option for each of the followings :

[10]

- 1 Let $A = \{6, 7, 8, 9, 10\}$ $B = \{2, 3, 6, 9, 11\}$ then $(A \cap B) =$ _____
(a) $\{6, 9\}$ (b) $\{2, 3, 6\}$ (c) $\{6, 7, 9\}$ (d) $\{6\}$
- 2 $A \cup U =$ _____
(a) U (b) $\emptyset$ (c) A (d) none
- 3 A function $f: A \rightarrow B$ is said to be _____ if each element of B is the image of some element of A
(a) onto (b) one to one (c) invertible (d) none
- 4 Let $A = \{n: n \text{ is odd}\}$, Then A is closed under _____
(a) multiplication (b) addition (c) both (d) none
- 5 Which of the following is unity element of ring Z
(a) 0 (b) 1 (c) both (d) none
- 6 $A = (a_{ij})$ is upper triangular matrix if its elements _____ are all zero
(a) above the diagonal (b) below the diagonal
(c) (a) & (b) both (d) none
- 7 Let $u = (2, -7, 1)$ $v = (-3, 0, 4)$ then $u + v =$ _____
(a) $(-1, -7, 5)$ (b) $(5, 7, 5)$ (c) $(1, 7, 5)$ (d) none
- 8 The observation whose frequency is highest is called _____
(a) mean (b) median (c) mode (d) none
- 9 Median of 3, 7, 11, 9, 13, 1, 7, 12, 18, 6, is
(a) 9 (b) 7 (c) 8.5 (d) none
- 10 The mean of first 10 natural numbers is
(a) 5 (b) 5.5 (c) 6.5 (d) 1

Q-2 Do as Directed (any 10)

[20]

- 1 Write down power set of $\{4, 6\}$
- 2 State the principle of mathematical induction in two equivalent forms.
- 3 If $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{4, 5, 6, 7, 8, 9\}$ and $C = \{2, 3, 5, 7, 8, 9\}$,
Find $(A \cap B) \oplus (B \cap C)$
- 4 Let $B = \{1, 2\}$. Is B closed under multiplication and addition.
- 5 For rational number a and b Define $a * b = ab/2$. Find identity element for given binary operation $*$.
- 6 Show that right cancellation law holds in group G .
- 7 If A and B are symmetric then, Is it true that AB symmetric? Justify your answer

C.P.T.O.)

- 8 Find x if $B = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is skew symmetric
- 9 The following is the frequency distribution of the number telephone calls received in 245 successive one minute intervals at an exchange: *Find mean.*
- | | | | | | | | | |
|--------------|----|----|----|----|----|----|----|----|
| No. of calls | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Frequency | 14 | 21 | 25 | 43 | 51 | 40 | 39 | 12 |
- 10 Find the Geometric mean of 2, 4, 8, 12, 16, 24.
- 11 Prove that with the help of Algebra of sets : $A \cap (A \cup B) = A$
- 12 Find $\|u\|$ if $u = (3, -12, -4)$

Q-3 A Define symmetric difference of sets A and B and Also prove that $A \oplus B = (A/B) \cup (B/A)$ [05]

B Prove that $p(n) : 1+2+3+\dots+n = n(n+1)/2$ [05]

OR

Q-3 A If Q is defined by $Q(a,b) = \begin{cases} 0 & \text{if } a < b \\ Q(a-b, b) + 1 & \text{if } a \geq b \end{cases}$ [05]
Then find $Q(2,3)$, $Q(19,5)$

B prove that with the help of laws of Algebra of sets:
(i) $(\emptyset \cup A) \cap (B \cup A) = A$ (ii) $(B^c \cap U) \cap (A^c \cup \emptyset) = (A \cup B)^c$ [05]

Q-4 A Consider the set Z = the set of integers. Which of the following operations in Z are associative: (i) addition (ii) multiplication (iii) subtraction (iv) division. Justify your answers [05]

B Define: (i) commutative ring (ii) unit element in a ring
(iii) $(a^{-1})^{-1} = a$, for any element a in G . [05]

OR

Q-4 A Define $*$ on Q by $a*b = a+b-ab$. [05]

(i) Is $(Q, *)$ semigroup?

(ii) Is $(Q, *)$ monoid?

(iii) find the inverses of elements of $(Q, *)$, if exist

B Let $f(x) = 2x^2 + 4x + 4$ and $g(x) = 3x^3 + 6x + 3$ Find roots of $f(x)$ over Z_{10} and Find roots of $g(x)$ over Z_3 . [05]

Q-5 A If $A = \begin{bmatrix} 2 & 6 & 2 \\ 5 & -1 & 3 \\ 3 & 1 & -2 \end{bmatrix}$ then find $A.A^T$ [05]

B Solve by determinants : $\begin{cases} 2x - 4y = 7 \\ 3x - 6y = 5 \end{cases}$ [05]

OR

Q-5 A Let $u = (5, 4, 1)$ $v = (-3, 0, 4)$ and $w = (0, 5, -8)$ and $k = 2$ find (i) $5u + 2v + 3w$ (ii) $\|w\|$ [05]
(iii) $(k \cdot v) + w$ (iv) which of the two vectors are orthogonal.

B If $A = \begin{bmatrix} 1 & -2 & 5 \\ -4 & -3 & 6 \\ 1 & 5 & 7 \end{bmatrix}$ $B = \begin{bmatrix} 2 & -2 & 0 \\ 0 & 4 & -5 \\ 2 & 6 & -4 \end{bmatrix}$ then find $3A + 2B$ [05]

Q-6A For the given data, find mean, harmonic mean and geometric mean. [05]

x_i	10	20	30	40	50	60	70
f_i	7	20	43	58	35	18	4

B Calculate Harmonic mean, Median, Mode for the following data. [05]

Marks	20	30	36	40	50
No. of students	2	7	10	5	1

OR

Q-6A Calculate Mean, Median and Mode for the following data. [05]

Weight (lbs) X	130	135	140	145	146	148	149	150	157
No. of persons	3	4	6	6	3	55	2	1	1

B From the following data calculate the missing value when Mean is 115.86 [05]

X	110	112	113	117	?	125	128	130
f	25	17	13	15	14	8	6	2

—X—

